

Zachary Hervieux-Moore

Profile

Applied scientist working on sequential decision making: reinforcement learning, bandits, and large-scale optimization. I have taken methods from research through production across retail, robotics, and cloud, and have led teams of scientists and engineers doing the same.

Contact

Email: ztjh@alumni.princeton.edu
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Website: hervature.com

Skills

Programming:
Python, C++

Machine Learning:
PyTorch, Jax, Ray, scikit-learn

Methods:
Deep RL, bandits, Bayesian optimization, transformers

Cloud:
AWS — SageMaker, Batch, ParallelCluster, EC2, S3

Languages:
English, Spanish, French

Education

Princeton University 2016-2021

- PhD & Master's, Operations Research and Financial Engineering
- Thesis: Modern Reinforcement Learning Techniques to Deal with Large Action Spaces

Queen's University 2011-2016

- Bachelor of Applied Science, Mathematics and Engineering, Systems and Robotics Option
- Bachelor of Arts, Economics

Awards

- Princeton SEAS Excellence in Teaching Award, 2020 & 2021 (student-selected)

Experience

Amazon, Las Vegas, NV
Senior Applied Scientist

2022-Present

- Author of more than 10 internal publications, including two Amazon Machine Learning Conference papers, two Amazon Fulfillment Simulation Symposium talks, and two Amazon Robotics Conference posters.

AWS — imPACT360

2026-Present

- Built a statistical engine that ranks a **22,000-account portfolio** for business development prioritization, drawing on robust trend estimation, change-point detection, parametric seasonality models, and calibrated win rates.
- Advancing the use of reinforcement learning for decision problems across AWS, and advising two junior scientists.

Amazon Robotics — Reinforcement Learning Group Lead

2024-2026

- Led a team of 8 scientists and engineers applying reinforcement learning and optimization to the robotic fulfillment network.
- Developed a combinatorial Bayesian optimization method that redesigns fulfillment center layouts with high-fidelity robotics simulators. The approach recovers **up to 5% additional storage density at constant throughput**, worth a **nine-figure annual saving** at network scale, and is deployed in production.
- Automated a layout design process that previously took expert engineers several weeks, producing a higher-performing design in **under a day**.

Corporate Projects — Reinforcement Learning Group Lead

2022-2024

- Co-founded and led a team of 8 scientists and engineers** delivering reinforcement learning solutions across the company; hired the team and mentored its scientists while remaining its lead individual contributor.
- Created a deep discrete choice bandit that selects Amazon's ~20M-product core assortment from a 500M+ item catalog; live experiments showed **annualized revenue lift in the tens of millions of dollars**.
- Designed a transformer mixture-of-experts recommendation model for sub-same-day delivery that achieved the **highest win rate (40%) among 70+ competing strategies** in a live test; selected for Amazon-wide deployment.
- Built the team's distributed reinforcement learning platform on AWS with Ray and RLlib, supporting training from a single workstation to several hundred GPUs, and used it to train PPO policies for robotic bin packing.

Harvard University, Cambridge, MA
Postdoctoral Researcher

2021-2022

- With Francesca Dominici, built deep learning models that estimate daily nationwide wildfire smoke exposure from satellite imagery; this work led to a publication in *Nature Human Behaviour*.
- With Natesh Pillai, developed a reinforcement learning method for learning improved MCMC samplers.

Correlation One, NYC, NY
Data Scientist

2019-2021

- Developed and taught machine learning, deep learning, and reinforcement learning curriculum to **over 10,000 students** across 8 editions of the Data Science for All program; lead instructor for two Amazon corporate training cohorts.

Stratify, NYC, NY
Data Scientist

2018-2019

- Developed a discretization algorithm that lifted the company's interpretable credit risk models to state-of-the-art performance.